

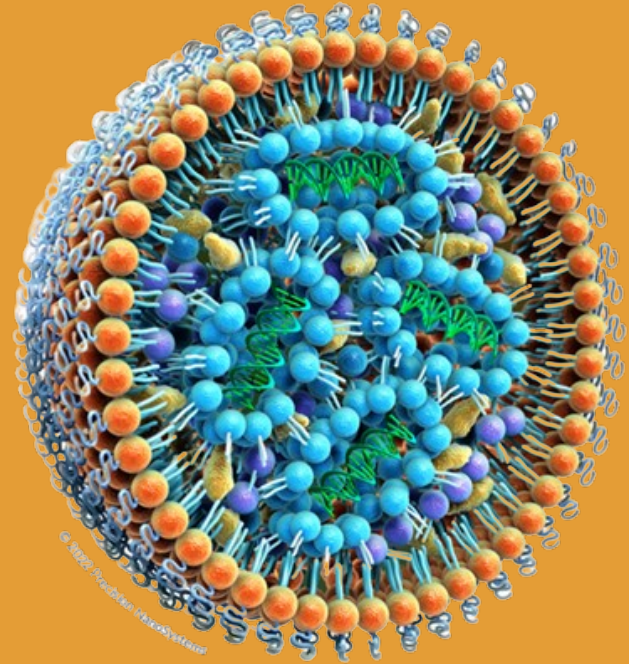
Lipid Nanoparticles: Helper Lipid Applications

Abigail Hartzell
Instructor: Dr. Ying Liu



What is an LNP?

Lipid nanoparticles: spherical vesicles that are composed of four specific types of lipids



LNP Structure [7]

How do Nucleic Acid-LNPs work?

- LNPs mimic lipoproteins that of low-density, allowing them to be taken up by an endogenous cellular transport pathway to deliver the nucleic acids to cells
- pH-sensitivity of the lipids allows the nucleic acids to be released from the LNPs into the cytoplasm when the pH of the vesicle decreases

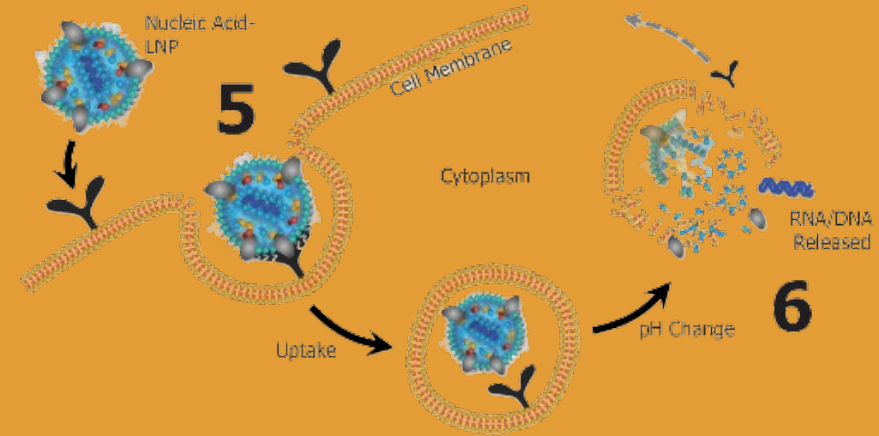
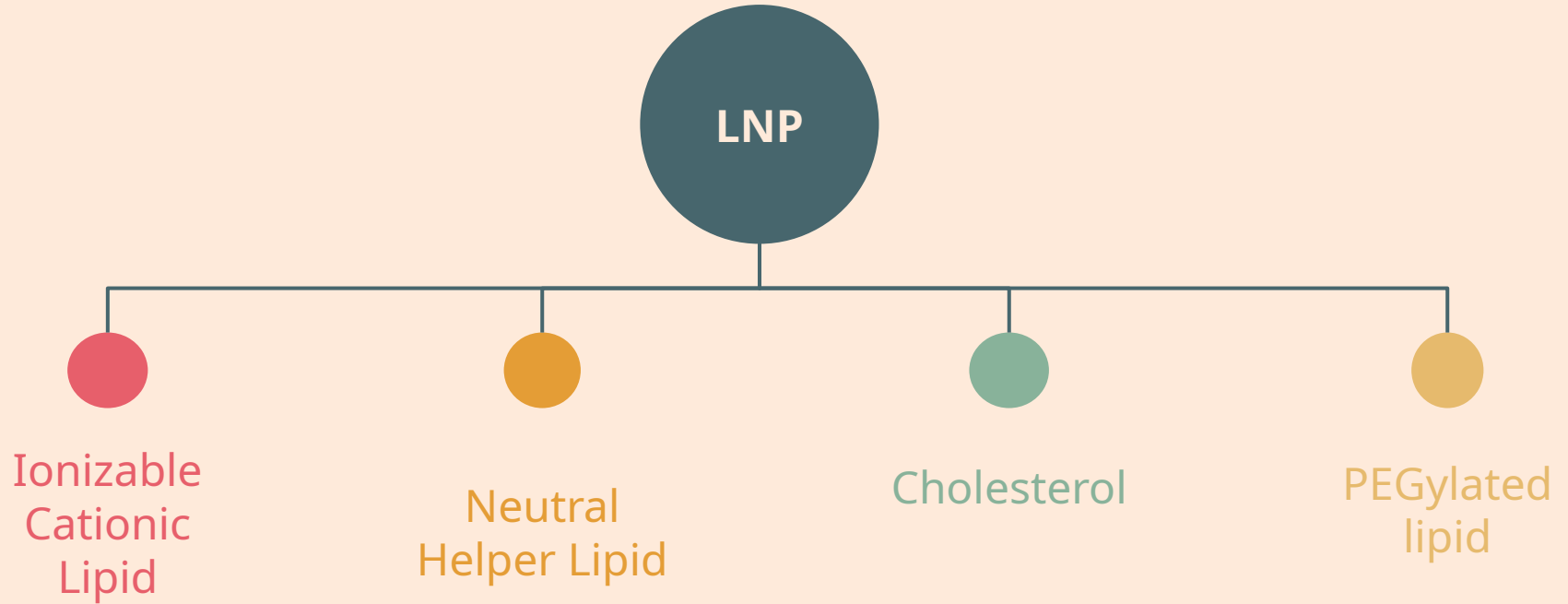


Figure 2. LNP Process [7]

Lipids Within an LNP



Lipids Within an LNP [1, 2, 3]

1

Ionizable Cationic Lipid

- Protect RNAs and facilitate transport
- Benefits of functioning due to pH
- 5 types: unsaturated, multi-tail, polymeric, biodegradable, branched-tail

2

Neutral Helper Lipid

- Provide support to LNPs
- Examples: 1,2-Dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE) and Phosphatidylcholine (PC)

3

Cholesterol

- Used for stability of lipid bilayers
- Can also provide better support and stability in lipid bilayers

4

PEGylated Lipid

- Lipid with strands of polyethylene glycol (PEG) attached
- Can enhance stability and increase transport
- Example: mPEG-DSPE

LNP Stability [6]

- Gel electrophoresis, fluorescence correlation spectroscopy (FCS), HPLC
- Chemical instability → hydrolysis and oxidation
- Physical instability

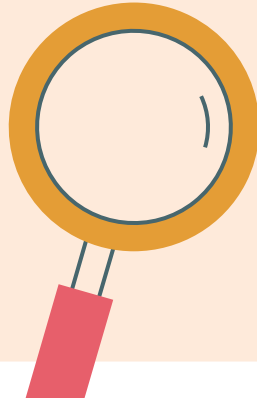
- mRNA Stability
 - mRNA may limit the stability instead of the actual LNPs
 - More stability with siRNA
- LNPs susceptible to hydrolysis due to aqueous environment
- mRNA can be stored for a few days at 4°C

LNP Storage

- Long-term storage have negative effects on efficacy and stability over time
- Low temperature storage
- pH of storage does not affect
- Freeze-thaw cycles

HYPOTHESIS:

Helper lipids in lipid nanoparticles can be optimized to allow for delivery of nucleic acids primarily to the lungs.



Why is this important?

Current Delivery

- Liver delivery
- Could expand delivery destinations

Merit

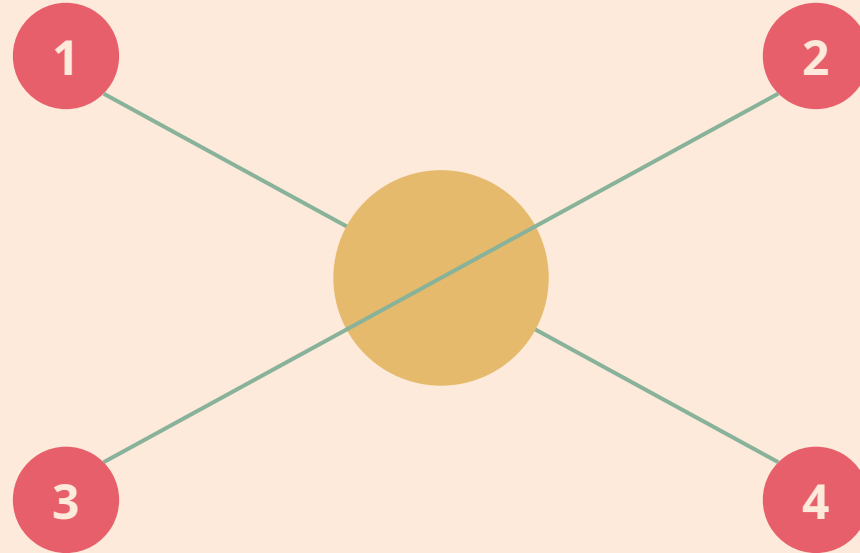
- Other advantages on stability

Lung Diseases

- Cystic fibrosis
- Alpha-1 antitrypsin deficiency (AATD)

Considerations

- Storage and administration

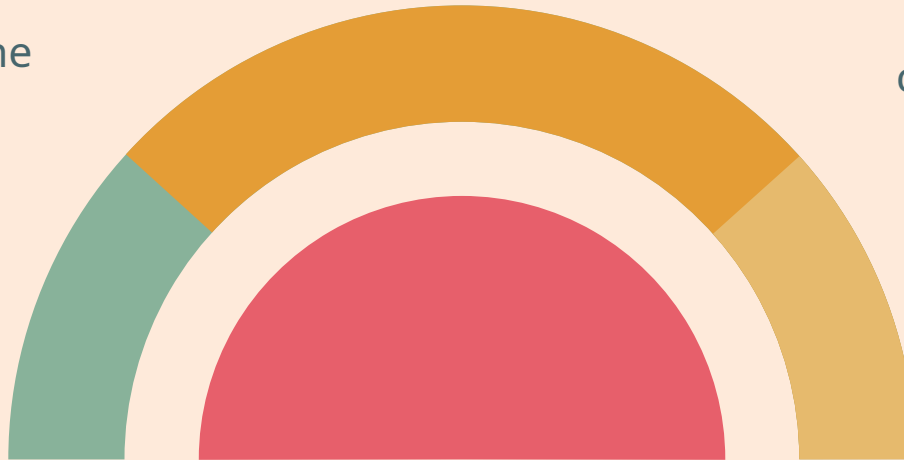


How can this be tested?

Lipid Choices
Combinations of
the lipids

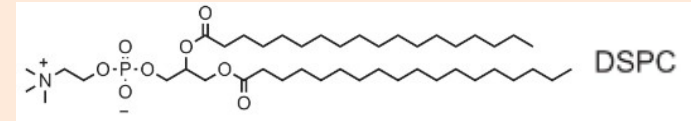
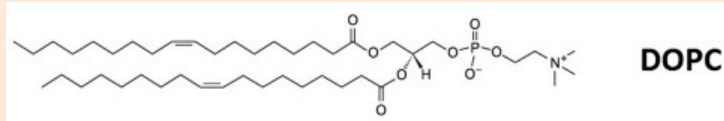
Structures
Branching in the
helper lipids

Composition
Molar
compositions to
change



Helper Lipid Structures

- The various structures can have effects on the overall LNPs:
 - Particle size
 - Charge
 - Surface modification
 - Efficacy

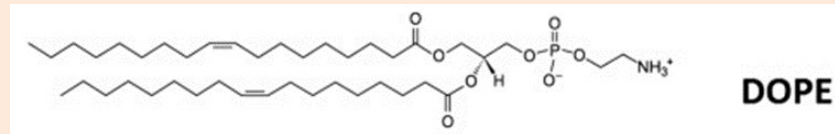


Molecular structure of common helper lipids, Dipalmitoylphosphatidylcholine (DOPC) and Distearoylphosphatidylcholine (DSPC) [11, 12]



Lipid Choices

- Keep the other 3 lipids the same
- Change the overall charge of the environment [8]
- Replace with:
 - Other neutral helper lipids: DOPC, sphingomyelin, and ceramide
 - Anionic lipids: phosphatidylserine (PS), phosphatidylglycerol, and phosphatidic acid
 - Cationic lipids: DOTAP and ethyl phosphatidylcholine



Molecular structure of the helper lipid 1,2-Dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE) [4]



Compositions of Lipids

- Current molar compositions used: 50:10:38.5:1.5 (ionizable lipid/DSPC/cholesterol/PEG2000-DMG) [9]
- Varying compositions of all lipids
 - Ionizable lipid: keep the same composition
 - Neutral helper lipid: vary this
 - Cholesterol: keep the same composition
 - PEGylated lipid: low (2.5, 6.5 and 15%) to high (25, 30, 35, 45, 55 and 60%) [10]



Applications

- Methods of testing:
 - Cell cultures to test in vitro first
 - Liver dissociation can be compared to lung dissociation in flow cytometry
- Cationic lipid replacement of the helper lipid shows potential
 - A study by LoPresti in the Journal of Controlled Release found that “replacing DOPE with DOTAP increased positive LNP surface charge at pH 7 by 5-fold and altered the ratio of liver to lung protein expression from 36:1 to 1:56” [8]
- Compositions need to be further looked into
 - A study by Lokugamage found that LNPs did not have a high enough efficacy to make conclusions for nebulized treatments of mRNA to the lungs [10]



Acknowledgement

This project is completed as part of the course CHE 422 at University of Illinois at Chicago under the guidance of Dr. Ying Liu.



References

1. Schoenmaker, Linde, et al. "mRNA-Lipid Nanoparticle COVID-19 Vaccines: Structure and Stability." *International Journal of Pharmaceutics*, vol. 601, 2021, pp. 120586–120586, <https://doi.org/10.1016/j.ijpharm.2021.120586>.
2. Mirjalili Mohanna, Seyedeh Zeinab, et al. "LNP-Mediated Delivery of CRISPR RNP for Wide-Spread in Vivo Genome Editing in Mouse Cornea." *Journal of Controlled Release*, vol. 350, 2022, pp. 401–13, <https://doi.org/10.1016/j.jconrel.2022.08.042>.
3. Han, Xuexiang, et al. "An Ionizable Lipid Toolbox for RNA Delivery." *Nature Communications*, vol. 12, no. 1, 13 Dec. 2021, [www.nature.com/articles/s41467-021-27493-0](https://doi.org/10.1038/s41467-021-27493-0), <https://doi.org/10.1038/s41467-021-27493-0>.
4. Cheng, Xinwei, and Robert J. Lee. "The Role of Helper Lipids in Lipid Nanoparticles (LNPs) Designed for Oligonucleotide Delivery." *Advanced Drug Delivery Reviews*, vol. 99, Apr. 2016, pp. 129–137, [www.sciencedirect.com/science/article/pii/S0169409X16300539?via%3Dihub](https://doi.org/10.1016/j.addr.2016.01.022), <https://doi.org/10.1016/j.addr.2016.01.022>.
5. Anzalone, Andrew V., et al. "Genome Editing with CRISPR–Cas Nucleases, Base Editors, Transposases and Prime Editors." *Nature Biotechnology*, vol. 38, no. 7, 22 June 2020, pp. 824–844, [www.nature.com/articles/s41587-020-0561-9](https://doi.org/10.1038/s41587-020-0561-9), <https://doi.org/10.1038/s41587-020-0561-9>.
6. Ball, Rebecca L., et al. "Achieving Long-Term Stability of Lipid Nanoparticles: Examining the Effect of pH, Temperature, and Lyophilization." *International Journal of Nanomedicine*, vol. 12, 2017, pp. 305–15, <https://doi.org/10.2147/IJN.S123062>.
7. "Lipid Nanoparticles." *Precisionnanosystems.com*, 2018, www.precisionnanosystems.com/workflows/formulations/lipid-nanoparticles.
8. LoPresti, Samuel T, et al. "The Replacement of Helper Lipids with Charged Alternatives in Lipid Nanoparticles Facilitates Targeted mRNA Delivery to the Spleen and Lungs." *Journal of Controlled Release*, vol. 345, 1 Mar. 2022, pp. 819–831, [www.sciencedirect.com/science/article/pii/S0168365922001808](https://doi.org/10.1016/j.jconrel.2022.03.046), <https://doi.org/10.1016/j.jconrel.2022.03.046>.
9. Wang, Wei, et al. "Prediction of Lipid Nanoparticles for mRNA Vaccines by the Machine Learning Algorithm." *Acta Pharmaceutica Sinica B*, vol. 12, no. 6, 2 Dec. 2021, pp. 2950–2962, [www.sciencedirect.com/science/article/pii/S2211383521004597](https://doi.org/10.1016/j.apsb.2021.11.021), <https://doi.org/10.1016/j.apsb.2021.11.021>.
10. Lokugamage, Melissa P, et al. "Optimization of Lipid Nanoparticles for the Delivery of Nebulized Therapeutic mRNA to the Lungs." *Nature Biomedical Engineering*, vol. 5, no. 9, 1 Sept. 2021, pp. 1059–1068, [www.nature.com/articles/s41551-021-00786-x](https://doi.org/10.1038/s41551-021-00786-x), <https://doi.org/10.1038/s41551-021-00786-x>.
11. Cheng, Xinwei, and Robert J Lee. "The Role of Helper Lipids in Lipid Nanoparticles (LNPs) Designed for Oligonucleotide Delivery." *Advanced Drug Delivery Reviews*, vol. 99, 1 Apr. 2016, pp. 129–137, [www.sciencedirect.com/science/article/pii/S0169409X16300539](https://doi.org/10.1016/j.addr.2016.01.022), <https://doi.org/10.1016/j.addr.2016.01.022>.
12. Cheng, Xinwei, and Robert J Lee. "The Role of Helper Lipids in Lipid Nanoparticles (LNPs) Designed for Oligonucleotide Delivery." *Advanced Drug Delivery Reviews*, vol. 99, 1 Apr. 2016, pp. 129–137, [www.sciencedirect.com/science/article/pii/S0169409X16300539](https://doi.org/10.1016/j.addr.2016.01.022), <https://doi.org/10.1016/j.addr.2016.01.022>.